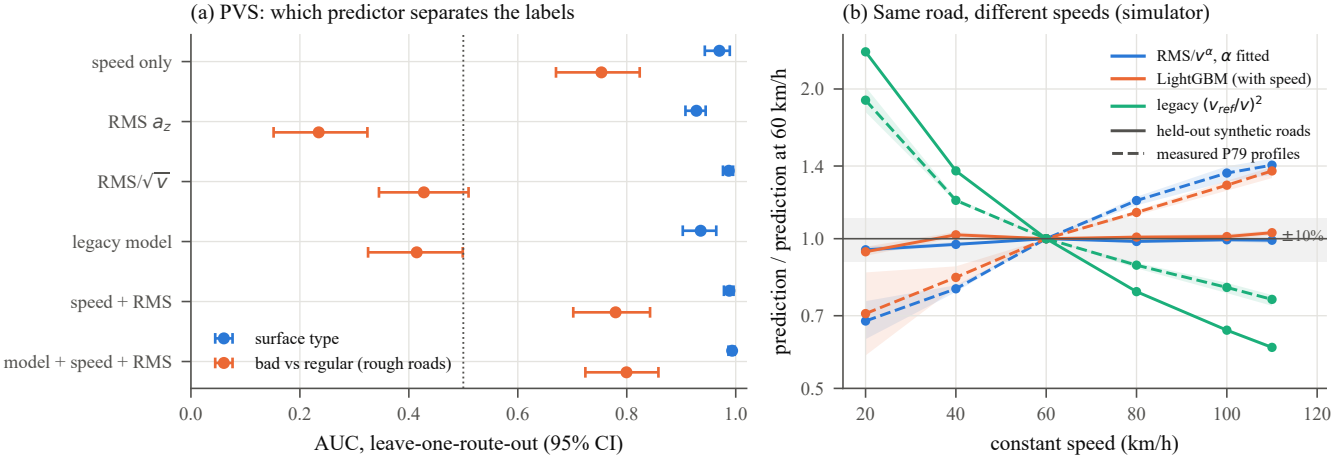
(a) PVS: which predictor separates the labels
speed only
RMS a_z
RMS/$\sqrt{v}$
legacy model
speed + RMS
model + speed + RMS
surface type
bad vs regular (rough roads)
0.0
0.2
0.4
0.6
0.8
1.0
AUC, leave-one-route-out (95% CI)
(b) Same road, different speeds (simulator)
RMS/v^{α}, α fitted
LightGBM (with speed)
legacy $(v_{ref}/v)^2$
held-out synthetic roads
measured P79 profiles
±10%
prediction / prediction at 60 km/h
2.0
1.4
1.0
0.7
0.5
20
40
60
80
100
120
constant speed (km/h)